Heavy-Tailed Return Modeling

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1 Introduction

Modeling the distribution of financial returns is a fundamental problem in quantitative finance. Asset-return distributions influence portfolio construction, derivative pricing, risk management, and stress testing. Many classical models, including Modern Portfolio Theory and Black-Scholes option pricing, rely either explicitly or implicitly on the assumption that returns are approximately Gaussian.

Under a Gaussian model, extreme market movements are exceedingly rare. For example, a daily return exceeding four standard deviations from the mean is expected to occur only a handful of times over several decades of trading. Empirical market data, however, often contradicts this prediction. Events commonly regarded as statistical outliers occur far more frequently than Gaussian theory would suggest.

This discrepancy has motivated extensive research into alternative distributional models capable of capturing the heavy-tailed nature of financial returns. Among these, the Student-t distribution has emerged as a widely used alternative due to its ability to model extreme observations while retaining analytical tractability.

This project investigates whether daily returns of the NIFTY 50 index can be adequately modeled using a Gaussian distribution. Using fifteen years of market data, we compare Gaussian and Student-t models through statistical diagnostics, likelihood-based estimation, and tail-event frequency analysis.

Core Idea

The central hypothesis of this study is that equity returns exhibit heavy-tailed behavior, causing extreme events to occur substantially more frequently than predicted under Gaussian assumptions.

2 Dataset and Preprocessing

Daily adjusted closing prices for the NIFTY 50 index were obtained through the Yahoo Finance API. The sample spans January 2010 through May 2025 and contains 3,781 trading-day observations. This period includes multiple market regimes, including the COVID-19 market shock of 2020, making it suitable for studying both typical and extreme return behavior.

Rather than analyzing raw prices, financial studies typically examine returns. Prices are non-stationary and trend over time, whereas returns provide a scale-independent measure of market movements that is more suitable for statistical analysis.

Log returns were computed as

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t denotes the adjusted closing price on day t.

Logarithmic returns possess several desirable properties. They are additive through time, approximately equal to arithmetic returns for small price movements, and arise naturally in many continuous-time financial models.

Scenario

Dataset: NIFTY 50 Index (`^NSEI`)
Period: Jan 2010 – May 2025
Observations: 3,781 daily returns

Even before formal testing, the empirical distribution exhibits fatter tails than a Gaussian density fitted to the sample moments, suggesting a higher frequency of extreme observations.

3 Distributional Diagnostics

The primary objective of this study is to evaluate whether Gaussian assumptions provide an adequate description of equity return distributions.

The following diagnostics were computed:

- Skewness
- Kurtosis
- Jarque-Bera Test
- Anderson-Darling Test

Metric	Value
Skewness	-0.906
Kurtosis	16.351
Jarque-Bera p-value	≈ 0
Anderson-Darling Statistic	33.10

A Gaussian distribution has kurtosis equal to 3. The observed kurtosis of 16.35 indicates substantially heavier tails and a higher frequency of extreme observations than predicted under normality.

Core Idea

The combination of elevated kurtosis and strong normality-test rejection suggests that extreme market movements are structural features of the data rather than rare anomalies.

Visual inspection provides a first indication that the Gaussian assumption may be inadequate. Figure 1 compares the empirical distribution of NIFTY returns against a Gaussian density fitted using the sample mean and variance. The Gaussian model captures the central region reasonably well but fails to account for the frequency of large positive and negative returns.

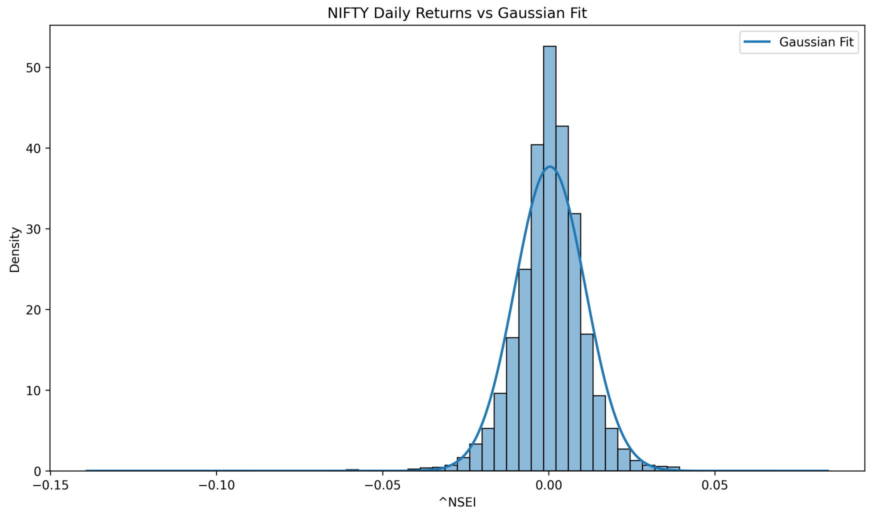

Figure 1: Distribution of daily NIFTY returns with fitted Gaussian density.

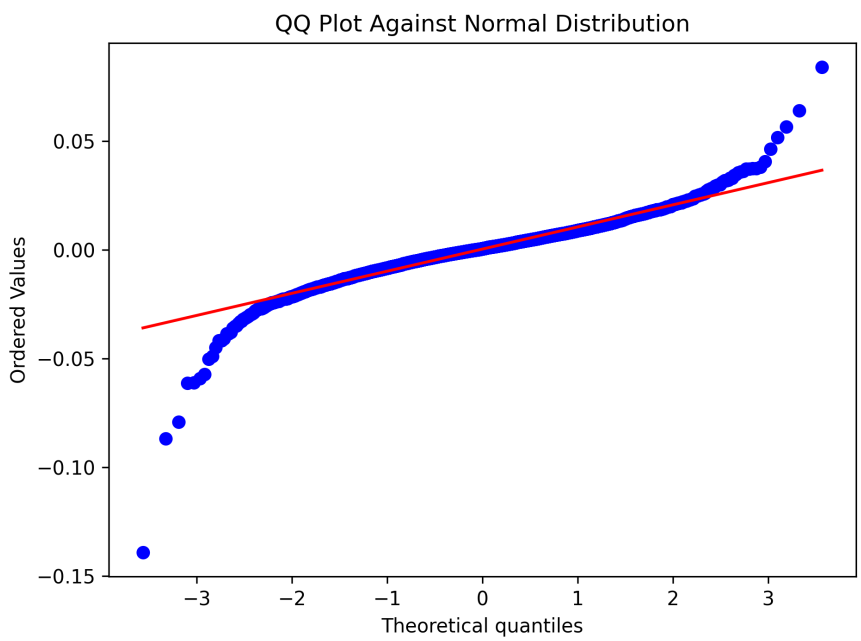

Figure 2: QQ plot against a Gaussian distribution.

The pronounced curvature in both tails indicates systematic deviations from Gaussian behavior. Extreme observations occur far more frequently than predicted under a normal model, providing strong visual evidence of heavy-tailed returns.

4 Parametric Distribution Fitting

To better characterize tail behavior, a Student-t distribution was fitted using Maximum Likelihood Estimation (MLE). Unlike the Gaussian distribution, the Student-t family includes a degrees-of-freedom parameter ν which controls tail thickness. As $\nu \to \infty$ the Student-t distribution converges to a Gaussian distribution. MLE estimation yielded $\nu = 4.07$ indicating pronounced heavy-tailed behavior.

Values of ν substantially below 10 are generally associated with pronounced heavy-tailed behavior relative to the Gaussian distribution.

Core Idea

A degrees-of-freedom estimate near 4 implies substantially heavier tails than a Gaussian distribution and is frequently observed in empirical studies of financial returns.

4.1 Likelihood Comparison

Model fit was evaluated using log-likelihood.

Model	Log-Likelihood
Gaussian	11831.64
Student-t	12173.57
Improvement	+341.93

The Student-t model achieves a log-likelihood improvement of 341.93, indicating a substantially better fit to the observed return distribution. Given that the Student-t model introduces only one additional parameter, the improvement is economically and statistically meaningful. This improvement indicates that the additional flexibility provided by the tail-thickness parameter is strongly supported by the data.

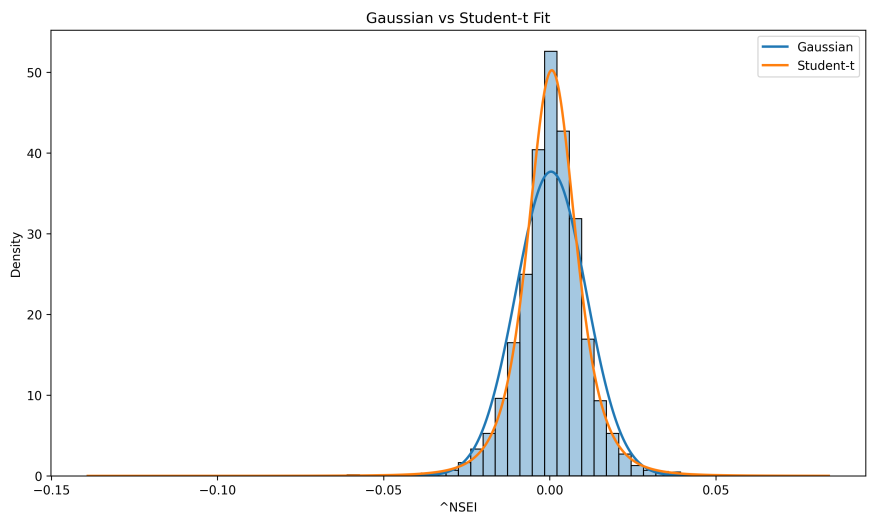

Figure 3: Gaussian versus Student-t distribution fit.

5 Tail Event Analysis

A key question in quantitative finance is whether observed extreme events occur at rates predicted by statistical models.

Standardized returns were computed as

$$z_t = \frac{r_t - \mu}{\sigma}$$

and tail-event frequencies were compared with Gaussian expectations.

Threshold	Observed	Expected	Ratio
3σ	43	10.21	4.21
4σ	15	0.24	62.63
5σ	10	≈ 0	Extremely Large

The divergence between empirical and Gaussian tail frequencies grows rapidly with increasing threshold levels, highlighting the inability of the normal distribution to capture extreme market movements. The increasing discrepancy provides direct empirical evidence that large market moves occur far more frequently than predicted under Gaussian assumptions.

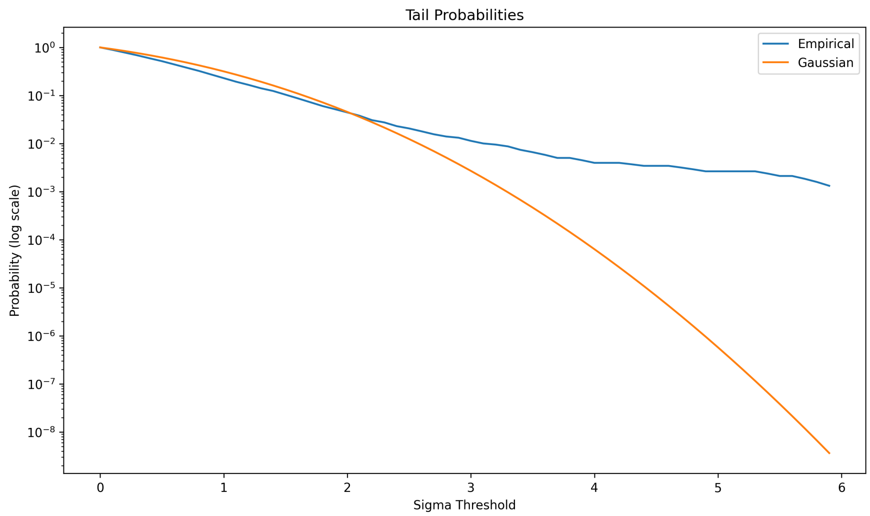

Figure 4: Empirical tail probabilities versus Gaussian predictions.

The gap between empirical and theoretical tail probabilities becomes increasingly pronounced beyond the 3σ threshold. This behavior is characteristic of heavy-tailed distributions and is inconsistent with Gaussian assumptions.

Scenario

A Gaussian model predicts approximately 0.24 observations exceeding 4 standard deviations in this sample. The dataset contains 15 such events, over 62 times the expected frequency.

6 Robustness Check

A common criticism of heavy-tail analyses is that the results may be driven entirely by a small number of extreme crisis observations. To test this, kurtosis was computed separately for crisis and non-crisis samples.

Period	Kurtosis
2020 Crisis Period	15.39
Non-Crisis Sample	5.47

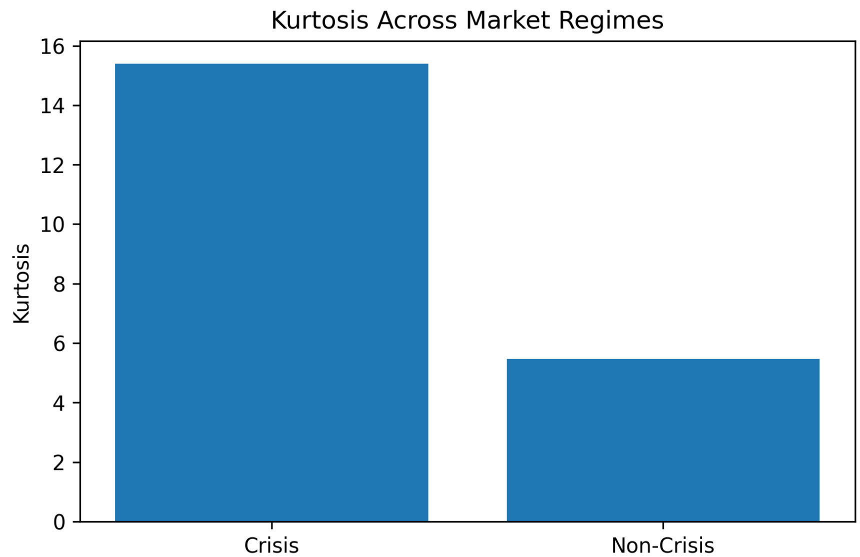

Figure 5: Kurtosis comparison across market regimes.

Although crisis periods amplify tail behavior, the non-crisis sample continues to exhibit substantial excess kurtosis. This indicates that heavy tails are a persistent feature of market returns rather than a consequence of isolated stress events.

Key Insight

A non-crisis kurtosis of 5.47 remains significantly above the Gaussian benchmark of 3, indicating that heavy tails are a persistent market feature rather than an artifact of a single crisis.

7 Conclusion

This study evaluated the suitability of Gaussian and Student-t models for describing the distribution of NIFTY 50 returns over a fifteen-year period. The evidence strongly rejects the Gaussian assumption for daily equity returns. Statistical tests, tail-frequency analysis, and likelihood-based model comparison all indicate the presence of substantial heavy-tailed behavior.

Key findings include:

- Normality is decisively rejected using both Jarque-Bera and Anderson-Darling tests.
- Return distributions exhibit substantial kurtosis and pronounced tail heaviness.
- A Student-t model with $\nu = 4.07$ provides a significantly better fit than a Gaussian distribution.
- Extreme 4σ events occur more than 62 times as frequently as predicted under normal assumptions.
- Heavy-tailed behavior persists even outside crisis periods.

The results consistently indicate that Gaussian assumptions underestimate the probability of extreme market movements. A Student-t specification with $\nu = 4.07$ provides a substantially better description of observed return behavior, highlighting the importance of heavy-tailed models in quantitative finance, risk management and derivative pricing.